

# Why Multimarket Banks Exist: A Quantitative Model With Demand Complementarity

DSE2026 Conference—Dynamic Finance and Banking

Chuqing Jin, Jack Liebersohn, and Yufeng Wu  
Toulouse, UC Irvine, and OSU

08/2026

# Consumers Prefer a One-stop Shop for Financial Products



 **Wealth Management**  
by Cerulli

[SUBSCRIBE TODAY](#) [Sign in](#)

[News](#) [Investment](#) [Wealth Planning](#) [Business Planning](#) [WealthTech](#) [RIA Edge](#) [Trusts & Estates](#) [Events & Awards](#)

## Clients Want One-Stop-Shop Banking: Cerulli Report

A survey showed that 66% of investors age 30 or younger prefer to use a single company for all their financial needs.

 **Michael Thrasher**, Staff Writer, WealthManagement.com  
June 19, 2019

 2 Min Read



### Latest news

**SUPPORT PLATFORMS**  
VRGL Expands Advisory Tech Platform With Growth Tools

**RIA**  
Thrivent Plans Another Hiring Spree for 2026

**RIA**  
Panel: Wealth Shops Need Ethical AI

# Consumers Prefer a One-stop Shop for Financial Products

Navigation: News Investment Wealth

Logo: Wealth Management

Buttons: SUBSCRIBE TODAY Sign in

Section: PYMNTS

Sub-sections: PYMNTS TV Today B2B Retail Fintech Digital Transformation Crypto AI PYMNTS® Int

## Clients Want C

A survey showed that 66% of

Michael Thrasher, Staff Writer  
June 19, 2019

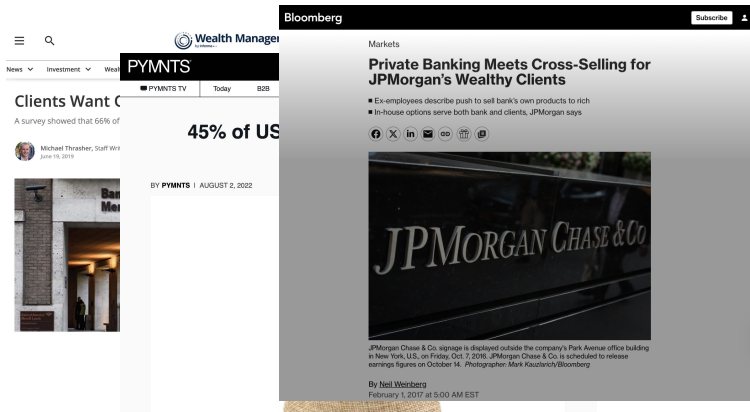
### 45% of US Consumers Want Banking Bundles

BY PYMNTS | AUGUST 2, 2022



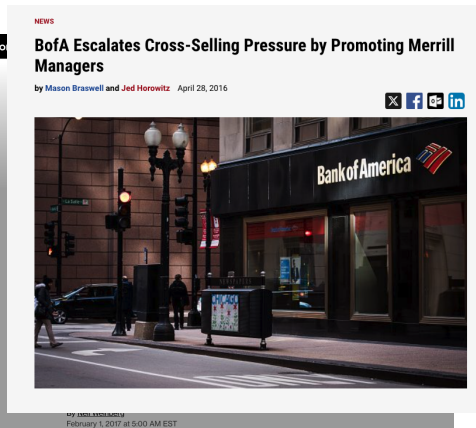
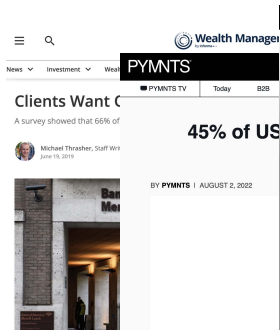
# Consumers Prefer a One-stop Shop for Financial Products

- It presents an important strategic priority for banks.



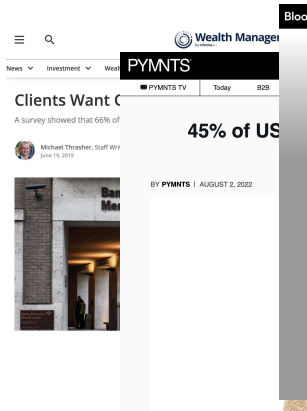
# Consumers Prefer a One-stop Shop for Financial Products

- It presents an important strategic priority for banks.



# Consumers Prefer a One-stop Shop for Financial Products

- It presents an important strategic priority for banks.



## BofA Escalates Wealth Managers

by Mason Braswell



NEWS

## Wells Fargo, BofA Execs Spotlight Cross-Selling Opportunities

by Miriam Rozen September 9, 2025



Wells Fargo and Bank of America executives reiterated this week the value they put on wealth advisors cross-selling consumer banking services.

# Consumers Prefer a One-stop Shop for Financial Products

- ▶ ... they are leveraging consumers' “complementary preference” in their demand
  - customers who hold one product often derive **additional convenience and value** from related services within the same institution
- ▶ Despite widespread anecdotes, such preference (and its implication for banks) remains underexplored in finance

# Why Not?



# Why Not?

Two challenges:

- ▶ Identification: contamination from correlated preferences
  - some unobservable factors drive consumers to deposit and borrow from the same institution
- ▶ Quantification: how banks endogenously react to such preference
  - these long-term, equilibrium effects are difficult to identify via reduced-form strategies

# What Do We Do?

Propose two solutions:

1. Propose an identification strategy
  - exploit local shift in Conforming Loan Limit (CLL) and banks' varying exposure to it
  - examine the elasticity of deposit with respect to mortgage supply

# What Do We Do?

Propose two solutions:

1. Propose an identification strategy
    - exploit local shift in Conforming Loan Limit (CLL) and banks' varying exposure to it
    - examine the elasticity of deposit with respect to mortgage supply
  2. Introduce a structural banking model
    - endogenize banks' responses and strategic interactions
- We use (2) to match the effect in (1), backing out the underlying demand complementarity
- Perform counterfactuals on (2) to explore the implications of cross-selling on: banks' value, cross-market pricing strategies, and interconnectedness

# What Do We Find

- ▶ Substantial demand complementarity
  - a mortgage relationship raises the probability of choosing the same bank for deposits by 40%.
- ▶ We compare a counterfactual model where we eliminate such demand complementarity, and thus banks' ability to cross-sell, with the baseline model

## Implication: the Value of Cross-selling

- ▶ Banks leverage their ability to cross-sell to these customers:
  - which generate 2.5% of market value
  - 5.4% of the deposit franchise and 13.4% of the mortgage business.

# Implication: the Value of Cross-selling

- ▶ Banks leverage their ability to cross-sell to these customers:
  - which generate 2.5% of market value
  - 5.4% of the deposit franchise and 13.4% of the mortgage business.
- ▶ Sources of economies of scope differ from those in the literature
  - mitigate moral hazard problems (Diamond and Rajan, 2000)
  - manage financial costs associated with liquidity risks (Kashyap, Rajan, and Stein, 2002; Gatev, Schuermann, and Strahan, 2009) and collaterals (Piazzesi and Schneider, 2020)

## Implication: Cross-market Pricing Strategies

- ▶ Give rise to higher market power in both deposit and mortgage markets
- ▶ Banks tilt their relative pricing across markets
  - larger increase in markup in the deposit market, where consumers are less rate sensitive
  - smaller increase in markup in mortgage market, as a way to expand their customer base

# Implication: Cross-market Pricing Strategies

- ▶ Give rise to higher market power in both deposit and mortgage markets
- ▶ Banks tilt their relative pricing across markets
  - larger increase in markup in the deposit market, where consumers are less rate sensitive
  - smaller increase in markup in mortgage market, as a way to expand their customer base
- ▶ Causing a redistributive effect different from what's emphasized in the prior literature, e.g., Park and Pennacchi (2008)
  - when banks face funding constraints
  - they redistribute toward the market that allows them to ease the constraint



## Implication: Interconnectedness across Markets

- ▶ Shocks to one market propagate to the other through demand complementarity
  - Simulate a shock to banks' mortgage market share
  - When banks can cross-sell, their depository activities are more responsive

# Implication: Interconnectedness across Markets

- ▶ Shocks to one market propagate to the other through demand complementarity
  - Simulate a shock to banks' mortgage market share
  - When banks can cross-sell, their depository activities are more responsive
- ▶ Quantitatively, the effect is comparable to traditional supply-side frictions emphasized in the literature (e.g., Gilje, Loutskina, and Strahan, 2016; Wang, Whited, Wu, and Xiao, 2022)
  - Funding constraints
  - Capital regulation

# Literature

- ▶ Banks' scope and interconnectedness
- ▶ Measure preference for financial services (Dick (2007), Ho and Ishii (2011), Buchak, Matvos, Piskorski, and Seru (2018), Koont (2025), Sarkisyan (2025).
  - focus on characteristics outside of the focal market
- ▶ Demand complementarity for goods Thomassen, Smith, Seiler, and Schiraldi (2017), Gentzkow (2007), Crawford (2008).
  - financial service with long-lasting (persistent relationships)
- ▶ Methodologically:
  - Dynamic banking models. Corbae and D'Erasmus (2021), Begenau and Landvoigt (2022), Dempsey (2025)
  - Using IVs to identify structural models

# Empirical Evidence on Demand Complementarity

- Why a naive approach does not work
- What do we do

# Why a Naive Approach Does Not Work?

A standard way of modeling (mortgage demand), following the long banking-IO literature: Buchak, Matvos, Piskorski & Seru, 2024, JPE; Jiang, 2023, RSF; Wang, Whited, Wu, & Xiao, 2022.

- ▶ Consumer  $i$  will choose the option that maximizes his (indirect) utility:

$$u_{i,j,t}^m = \beta_j^m \mathbf{x}_{j,t}^m - \alpha^m r_{j,t}^m + \xi_{j,t}^m + \epsilon_{i,j,t}^m$$

- mortgage rate
- other characteristics
- “quality shocks” at the bank-market level, unobserved to econometricians
- some other personal, idiosyncratic, “taste shocks” (following exogenous distribution)

## Why a Naive Approach Does Not Work?

A standard way of modeling (mortgage demand), following the long banking-IO literature:

- ▶ Consumer  $i$  will choose the option that maximizes his (indirect) utility:

$$u_{i,j,t}^m = \beta_j^m \mathbf{x}_{j,t}^m - \alpha^m r_{j,t}^m + \xi_{j,t}^m + \gamma \mathbb{I}_{i,j,t}^d + \epsilon_{i,j,t}^m$$

- mortgage rate
- other characteristics
- **do I currently deposit with the bank (demand complementarity)**
- “quality shocks” at the bank-market level, unobserved to econometricians
- some other personal, idiosyncratic, “taste shocks” (following exogenous distribution)

# Why a Naive Approach Does Not Work?

We model demand for deposits symmetrically:

- ▶ Consumer  $i$  will choose the option that maximizes his (indirect) utility:

$$u_{i,j,t}^d = \alpha^d r_{j,t}^d + \beta^d \mathbf{x}_{j,t}^d + \xi_{j,t}^d + \gamma \mathbb{I}_{i,j,t-1}^m + \epsilon_{i,j,t}^d$$

- deposit rate
- other characteristics
- **do I have a previous mortgage relationship with the bank?**
- “quality shocks” at the bank-market level, unobserved to econometricians
- some other personal, idiosyncratic, “taste shocks” (following exogenous distribution)

## Why a Naive Approach Does Not Work?

If we regress  $\Delta \log \text{Deposit}_{ib,t}$  on  $\Delta \log \text{MortOrig}_{ib,t}$

- ▶ unobserved “quality shock” might be correlated

Having account level data does NOT help

- ▶ idiosyncratic “taste shocks” may be correlated



# Why a Naive Approach Does Not Work?

If we regress  $\Delta \log \text{Deposit}_{ib,t}$  on  $\Delta \log \text{MortOrig}_{ib,t}$

- ▶ unobserved “quality shock” might be correlated

Having account level data does NOT help

- ▶ idiosyncratic “taste shocks” may be correlated

## What do we do

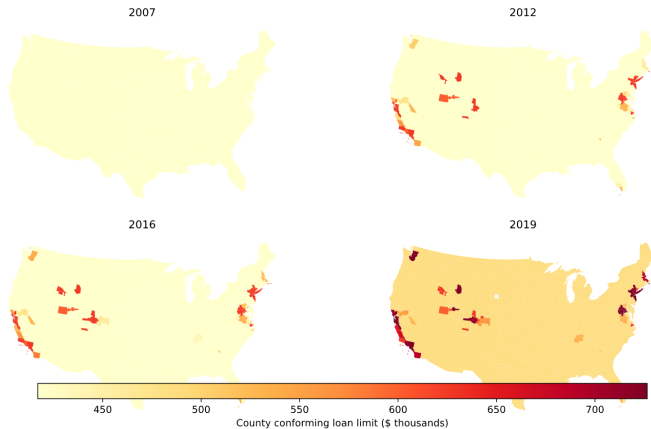
- ▶ How do plausibly exogenous shocks to mortgage market, driven by conforming loan limit (CLL) spill over to deposit intake?

# Conforming Loan Limit

- ▶ What are they?
  - Fannie Mae and Freddie Mac (the GSEs) guarantee and securitize qualified mortgages
  - These mortgages cannot exceed a maximum amount
- ▶ Who sets the limits?
  - The Federal Housing Finance Agency (FHFA)
- ▶ How are they set?
  - FHFA set a national limit every year prior to 2008
  - Certain high-cost counties have higher limits after 2009, based on the median housing price in that county

# Conforming Loan Limit Variations

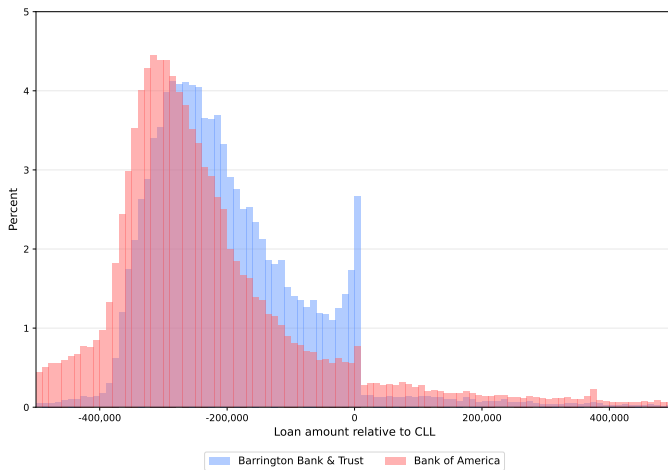
County-level conforming loan limits in selected years



# Empirical Strategy

- ▶ Regress changes in mortgages on changes in CLL\* banks' varying exposure to it
  - prior literature: has focused on averages
  - we focus on the redistributive effects
  
- ▶ Explore the fact that some banks are more sensitive to the changes than other banks

# Empirical Strategy



## Bartik Instrument: Exposure Share

**Bank exposure:** share just below the CLL minus share just above the CLL

- ▶ For bank  $i$  in county  $c$  and year  $t$ , compute

$$\text{Exposure}_{ict} = S_{ict}^{[0.95, 1.00)} - S_{ict}^{(1.00, 1.05]}$$

- ▶ Leave-one-out weighted average for bank  $i$ ,

$$\text{Exposure}_{id}^{(-d)} = \frac{\sum_{c \notin d, t} \text{Loan count}_{ict} \cdot \text{Exposure}_{ict}}{\sum_{c \notin d, t} \text{Loan count}_{ict}}$$

- the BoA v.s. Barrington bank idea
- higher exposure means the bank specializes more in conforming lending

## Bartik Instrument: Shift

**Bank exposure:** share just below the CLL minus share just above the CLL

**Local shift:** nearby loans that switch from jumbo to conforming when the CLL rises

$$\text{Shift}_{ibt}^{(-i)} = \frac{\text{Loans count near branch } b \text{ between old CLL and new CLL}}{\text{Total loan count near branch } b}, \text{ from other banks}$$

— *Leave-one-out* ratio near branch  $b$  of bank  $i$  in year  $t$

**Instrument:**  $Z_{ibt} = \text{Exposure}_{id}^{(-d)} \times \text{Shift}_{ibt}^{(-i)}$

## Local Projection

For branch  $b$  of bank  $i$  at horizon  $h$  from year  $t$ , estimation local projections

**First Stage:**  $\Delta \log \text{MortOrig}_{ib,t+h} = \theta_h Z_{ibt} + \Gamma_h X_{ibt} + \alpha_{it} + \delta_{ct} + u_{ib,t+h}$

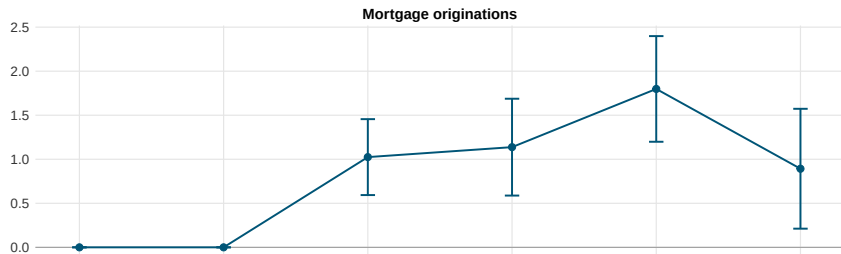
**Deposit Result:**  $\Delta \log \text{Deposit}_{ib,t+h} = \rho_h Z_{ibt} + \Pi_h X_{ibt} + \alpha_{it} + \delta_{ct} + v_{ib,t+h}$

- $\alpha_{it}$ : bank  $\times$  year FE
- $\delta_{ct}$ : county  $\times$  year FE
- $X_{ibt}$ : controls
- 5-mile radius around branch  $b$

**Identifying assumption:** within county-year and bank-year, conditional on controls, high-exposure banks would not have grown differently in high-shift local markets, relative to low-exposure banks, if the CLL had not changed.

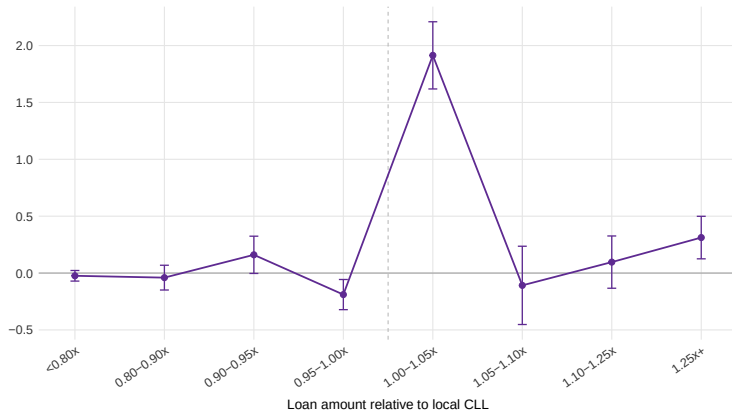


## Results on Mortgage Origination



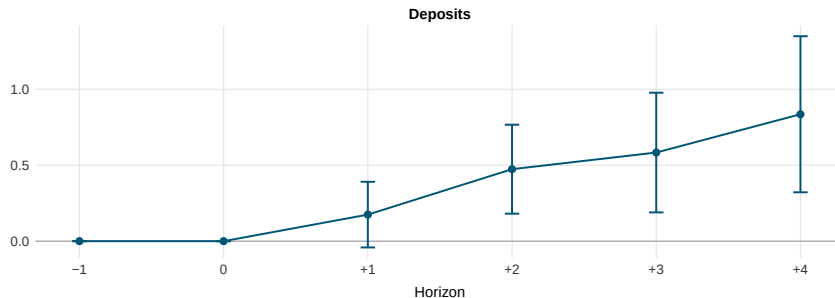
- change in mortgage growth for a one std change in the Bartik instrument
- the effect on mortgage picks up immediately

# Validating the Bartik Instrument



- response is localized to the CLL cutoff bucket
- no changes for loans further above or below the threshold

## Results on Deposits



- the change in deposits is gradual
- 5 cents growth in deposits for each dollar increase in mortgages

[Different Radius](#)[Pretrend](#)

# Dynamic Banking Model with Consumer Demand Complementarity

## Model Overview:

- ▶ Static players:
  - Mass W of depositors: choose which bank to deposit their money
  - Mass H of borrowers: choose which bank to take a mortgage
  - A shadow bank: chooses mortgage rate and securitizes everything
- ▶ **Dynamic players: N banks**, each
  - sets the deposit and mortgage rates
  - chooses to securitize a fraction of mortgages
  - chooses to invest in risky and risk-free assets
  - finances the asset side using equity, deposit, and wholesale funding
- ▶ In EQ, everyone rationally expects other players' decisions
  - Following the PBE definition in Weintraub, Benkard, & Van Roy (2008, 2010).

## Recap—How We Model Demand

**from mortgage borrowers:**

- ▶ Borrower  $i$  will choose the option that maximizes his utility:

$$u_{i,j,t}^m = \beta_j^m \mathbf{x}_{j,t}^m - \alpha^m r_{j,t}^m + \xi_{j,t}^m + \gamma \mathbb{I}_{i,j,t}^d + \epsilon_{i,j,t}^m$$

**from depositors:**

- ▶ Depositor  $i$  will choose the option that maximizes his utility:

$$u_{i,j,t}^d = \alpha^d r_{j,t}^d + \beta^d \mathbf{x}_{j,t}^d + \xi_{j,t}^d + \gamma \mathbb{I}_{i,j,t-1}^m + \epsilon_{i,j,t}^d$$

We will impose exogenous distribution (standard assumption, Type I Extreme Value), so that we can calculate aggregate market shares.

# The Resulting Mortgage Market Share:

$$\begin{aligned}
 s_{j,t}^{org} = & \underbrace{\frac{\exp(q_{j,t}^m - \alpha^m r_{j,t}^m + \gamma)}{\exp(q_N^m) + \exp(-\alpha^m r_{j+1,t}^m) + \exp(q_{j,t}^m - \alpha^m r_{j,t}^m + \gamma) + \sum_{h \in \mathcal{J} \setminus j} \exp(q_{h,t}^m - \alpha^m r_{h,t}^m)}}_{\text{market share among own depositors}} \cdot s_{j,t}^d \\
 & + \sum_{k \in \mathcal{J} \setminus j} \underbrace{\frac{\exp(q_{j,t}^m - \alpha^m r_{j,t}^m)}{\exp(q_N^m) + \exp(-\alpha^m r_{j+1,t}^m) + \exp(q_{k,t}^m - \alpha^m r_{k,t}^m + \gamma) + \sum_{h \in \mathcal{J} \setminus k} \exp(q_{h,t}^m - \alpha^m r_{h,t}^m)}}_{\text{market share among competitors' depositors}} \cdot s_{k,t}^d \\
 & + \underbrace{\frac{\exp(q_{j,t}^m - \alpha^m r_{j,t}^m)}{\exp(q_N^m) + \exp(-\alpha^m r_{j+1,t}^m) + \sum_{h \in \mathcal{J}} \exp(q_{h,t}^m - \alpha^m r_{h,t}^m)}}_{\text{market share among non-depositors}} \cdot \left( 1 - \sum_{k \in \mathcal{J}} s_{k,t}^d \right).
 \end{aligned}$$

- A bank's mortgage market share depends on its deposit market presence

# The Resulting Deposit Market Share:

$$\begin{aligned}
 s_{j,t}^d = & \underbrace{\frac{\exp(\alpha^d r_{j,t}^d + q_{j,t}^d + \gamma)}{1 + \exp(\alpha^d r_{j,t}^d + q_{j,t}^d + \gamma) + \sum_{h \in \mathcal{J} \setminus j} \exp(\alpha^d r_{h,t}^d + q_{h,t}^d)}}_{\text{market share among own mortgage customers}} \cdot s_{j,t-1}^m \\
 & + \sum_{k \in \mathcal{J} \setminus j} \underbrace{\frac{\exp(\alpha^d r_{j,t}^d + q_{j,t}^d)}{1 + \exp(\alpha^d r_{k,t}^d + q_{k,t}^d + \gamma) + \sum_{h \in \mathcal{J} \setminus k} \exp(\alpha^d r_{h,t}^d + q_{h,t}^d)}}_{\text{market share among competitors' mortgage customers}} \cdot s_{k,t-1}^m \\
 & + \underbrace{\frac{\exp(\alpha^d r_{j,t}^d + q_{j,t}^d)}{1 + \sum_{k \in \mathcal{J}} \exp(\alpha^d r_{k,t}^d + q_{k,t}^d)}}_{\text{market share among non-borrowers}} \cdot \left( 1 - \sum_{k \in \mathcal{J}} s_{k,t-1}^m \right).
 \end{aligned}$$

- ▶ A bank's deposit market share depends on its prior mortgage market presence
  - which evolves dynamically:  $s_t^m = (1 - \mu)s_{t-1}^m + \mu s_t^{org}$ , where  $\mu$  is the repayment maturity of mortgage



# Bank's Profit

Bank's balance sheet:

$$\underbrace{\text{Risky asset} + \text{Risk-free asset} + \text{Mortgage retained}}_{\text{Assets}} = \underbrace{\text{Deposit} + \text{Wholesale}}_{\text{Liabilities}} + \underbrace{K_{j,t}}_{\text{Equity}}$$

► Sources of profit:

- holding risky assets:  $\omega \cdot \text{Risky asset}^\theta$
- holding risk-free asset and get a pre-tax return of  $f$
- profit from the deposit market: PV of deposit spread – MC
- profit from the mortgage market: PV of future mortgage spread – MC

► Sources of expenses:

- (quadratic) financing cost of accessing wholesale funding
- cost of securitization  $\eta \times (s - \text{mortgage retained})$
- fixed operation cost  $\chi$

## Banks' Decision:

$$V^j(\omega_{j,t}, s_{j,t}^m, K_{j,t}) = \max \left\{ \text{Dividend} + \frac{1}{1+f} \left[ \mathbb{E} V^j(\omega_{j,t+1}, s_{j,t+1}^m, K_{j,t+1}) \right] \right\}$$

— s.t. the capital regulation:  $K_{j,t} \geq \kappa \times (\text{Risky asset} + \text{Mortgage retained})$

- ▶ The dynamic state variables of the bank:
  - $\omega$ : return on risky asset holding
  - $s$ : banks' mortgage base
  - $K$ : bank's equity capital:  $K_{j,t+1} = K_{j,t} + \text{Profit} - \text{Dividend}$
  
- ▶ Banks choose deposit and mortgage rates, the balance sheet variables (risky, risk-free investments, mortgage secularization, wholesale funding), and profit retention v.s. payout

## Estimation Strategy:

- ▶ Estimate standard household demand parameters (in banking IO)
  - Regress bank market share on deposit rates and bank characteristics
  - Instrument for deposit rates using standard cost shifters (e.g., salaries, noninterest expenses tied to fixed assets)
- ▶ Estimate bank cost parameters
  - Match moments of key balance sheet variables to pin down operating and funding costs
- ▶ Estimate **demand complementarity**
  - Match the results from the Bartik instrument
  - 1 dollar of mortgage translates into 5 cents of deposits

## Changes in CLL:

- ▶ Simulate a  $\Delta$  CLL shock in the model, we fix it to 1%
- ▶ Borrowers' utility  $u_{i,j,t}^m = \beta_j^m \mathbf{x}_{j,t}^m - \alpha^m r_{j,t}^m + \xi_{j,t}^m + \gamma \mathbb{I}_{i,j,t}^d + \epsilon_{i,j,t}^m$ 
  - households derive “conforming convenience” (Buchak et al. 2024, JPE)
  - banks are affected differentially (heterogeneous exposures to CLL).
- ▶ We calibrate  $\beta_{CLL,j}^m \times \Delta CLL$  to the distribution of the Bartik instrument in the data
- ▶ Calculate the same coefficient in the model simulated data

# Estimation Results

## Panel A: Demand Parameters

|                                   |                                          |        |
|-----------------------------------|------------------------------------------|--------|
| $\alpha^d$                        | Depositors' sensitivity to deposit rates | 0.831  |
| $\alpha^m$                        | Borrowers' sensitivity to mortgage rates | -1.372 |
| $\bar{q}_{j,j \in \mathcal{J}}^d$ | Mean quality of banks' deposits          | 2.308  |
| $\bar{q}_{J+1}^d$                 | Mean quality of holding MMF              | 0.010  |
| $\bar{q}_{j,j \in \mathcal{J}}^m$ | Mean quality of banks' mortgages         | -0.044 |
| $\gamma$                          | Demand complementarity                   | 0.414  |

## Panel B: Supply Parameters

|                 |                                                             |        |
|-----------------|-------------------------------------------------------------|--------|
| $c^d$           | Bank's marginal cost of taking deposits                     | 0.005  |
| $c^m$           | Bank's marginal cost of originating mortgages               | 0.001  |
| $\phi$          | Quadratic cost of wholesale funding                         | 0.005  |
| $c^s$           | Shadow bank's marginal cost of originating mortgages        | 0.007  |
| $\mu_\omega$    | Mean of log productivity from holding risky assets          | -3.291 |
| $\rho_\omega$   | Auto-correlation log productivity from holding risky assets | 0.675  |
| $\sigma_\omega$ | Std of log productivity from holding risky assets           | 0.186  |
| $\theta$        | Returns to scale from holding risky assets                  | 0.936  |
| $\chi$          | Fixed operating cost                                        | 0.177  |

## What does this imply for banks?

- compare with counterfactual models where we eliminate/vary demand complementarity

# Implications: The Value of Cross-selling

- ▶ Compare the baseline model and a counterfactual model WITHOUT complementary demand
- ▶ What is the value created?

|                   | Value component | % value of demand complementarity |
|-------------------|-----------------|-----------------------------------|
| Bank (total)      | 100.0%          | 2.49%                             |
| Deposit franchise | 45.87%          | 5.43%                             |
| Mortgage business | 18.55%          | 13.43%                            |

# Implications: The Value of Cross-selling

- ▶ Compare the baseline model and a counterfactual model WITHOUT complementary demand
- ▶ What is the value created?

|                   | Value component | % value of demand complementarity |
|-------------------|-----------------|-----------------------------------|
| Bank (total)      | 100.0%          | 2.49%                             |
| Deposit franchise | 45.87%          | 5.43%                             |
| Mortgage business | 18.55%          | 13.43%                            |

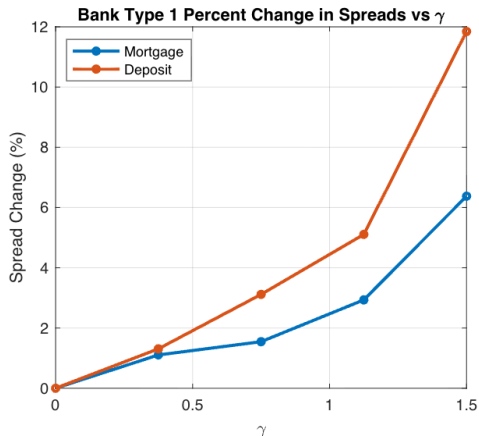
— economies of scope without suppliers' agency or financial frictions



## Implications: Cross-market Pricing

- ▶ How does the bank adjust its strategies in the deposit and mortgage markets?
- ▶ The answer is ex-ante unclear:
  - complementary demands give rise to market power
  - induces banks to compete more intensively for customer base
- ▶ Effect depends on the relative magnitudes of these forces

# Implications: Cross-market Pricing



- different from cross-subsidization of multi-market banks emphasized in the literature based on suppliers' (funding) constraints (e.g., Park and Pennacchi, 2008)

# Implications: Interconnectedness of Deposits and Mortgages

- ▶ From price to quantity: How does mortgage-market shock spill over to deposit intake?
- ▶ Counterfactual: 5 bps reduction in mortgage origination cost
  - Increased mortgage origination relative to baseline **1.98%**
  - Each increased mortgage dollar increases deposits by **7.2 cents**
- ▶ Shutoff demand complementarity:
  - Each increased mortgage dollar increases deposits by **2.2 cents**
  - Demand and Demand $\times$ Supply account for the rest

# Implications: Interconnectedness of Deposit and Mortgage Markets

- ▶ The effect capture very different concepts from the reduced-form spillover:
  - the macro as opposed to the micro elasticity
  - fully accounting for supply side adjustment
  - it measures effect in the steady state
- ▶ We estimate the model using cleanly identified local treatment effects  
then the estimated model to speak to more macro-level implications
- ▶ Quantify mechanisms and provide useful policy implications

## Implications: What Do Policy Makers Think?



On April 17, 2024, Governor Bowman participated in a discussion at the Institute for International Finance (IIF) Global Outlook Forum. (2024). [Photo]. Retrieved from [Federal Reserve](#).

Banks are originating a smaller portion of mortgages than they used to, and the Federal Reserve wants to change that with proposed rule shifts, according to one top regulator.

Fed Governor and Vice Chair of Supervision Michelle Bowman—

“Customers with strong bank connections naturally turn to that bank for other financial needs, from checking accounts to investment services. [...] good customer service in the mortgage business can lead to a stronger relationship with customers and result in improved financial resiliency of the banking sector”

—**the aggregate impact of “demand complementarity” matters**

# Conclusion

- ▶ Consumers exhibit **demand complementarity**
- ▶ A demand theory for binding depository and lending services

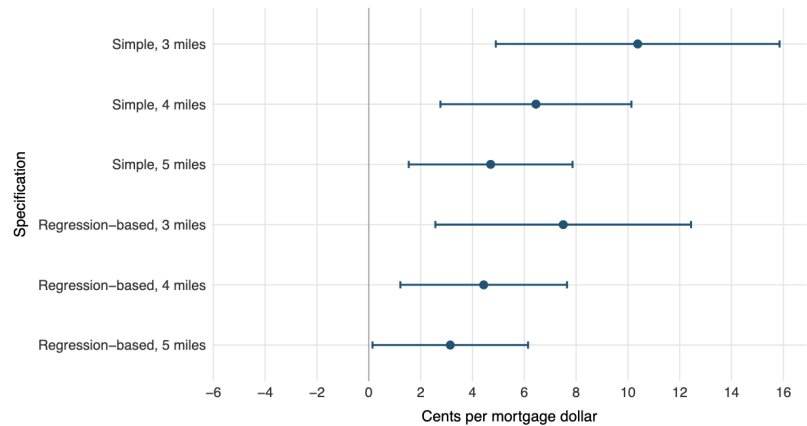
# Conclusion

- ▶ Consumers exhibit **demand complementarity**
- ▶ A demand theory for binding depository and lending services
- ▶ We focus on retail banking, but the mechanism is broader (e.g., investment banking, brokerage services).
  - especially after Basel III

THANK YOU!

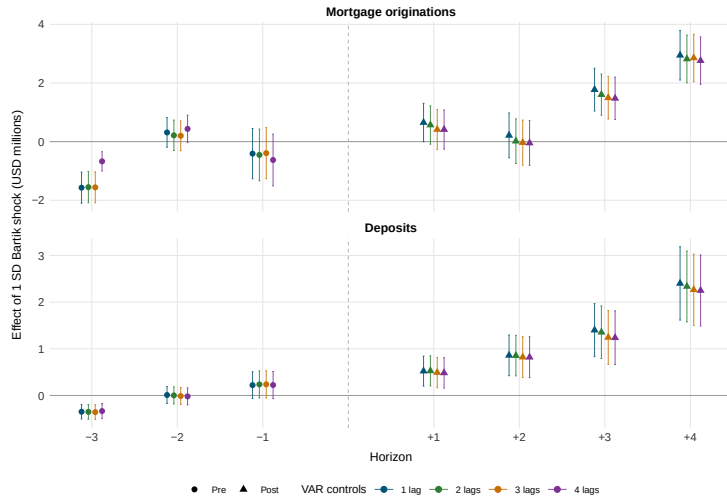


# Robustness



[Back to Results](#)

# Pre-trend Using National Only Sample



[Back to Results](#)